

AI61002 : Deep Learning Foundations and Applications

Spring 2025

Assignment 1 Outline: Neural Network for Regression

(20 Marks)

Due Date: February 2 (by 9 PM IST)

Instructions: Complete the tasks listed below, ensuring you follow the specified instructions for each. Use PyTorch for implementation and provide step-by-step outputs in the Jupyter Notebook (.ipynb file). All relevant code snippets, visualizations, and explanations must be included within the notebook. Submit the required files as per the submission guidelines provided.

Task 1 - 2 Mark

Split the given dataset into training, validation, and test sets in the ratio 70%-15%-15%. Ensure randomness in the split to avoid bias. Define corresponding dataloaders for each partition to facilitate batch-wise data loading.

Task 2 - 3 Marks

Design a fully connected neural network (FCNN) for regression tasks. Implement the following:

- Use ReLU activations for the hidden layers. (1 Mark)
- Decide the number of hidden layers and the units per layer through experimental cross-validation to achieve optimal performance. (1 Mark)
- Select an appropriate activation function for the output layer, considering the nature of the regression outputs. (1 Mark)

Task 3 - 1 Mark

Print the model summary, including the total number of trainable parameters and the model's computational requirements.

Task 4 - 2 Mark

Define an appropriate loss function based on the problem's requirements.

Task 5 - 3 Marks

Training Process:

Train the model using the following guidelines:

- Use the SGD optimizer with an initial learning rate. (1 Mark)
- Experiment with suitable batch sizes for efficient training.
- Implement learning rate scheduling to reduce the learning rate when the validation loss stagnates for a predefined number of epochs. (1 Mark)
- Track training and validation losses for every epoch and ensure overfitting is avoided.
- Clearly document the training process and any hyperparameter choices.

Task 6 - 2 Marks

Visualize the training and validation losses dynamically during the training process using Tensorboard.

Task 7 - 2 Marks

Save the trained model to a file for future use.

Task 8 - 2 Mark

Load the trained model and perform predictions on the test partition. Compute and report the Mean Squared Error (MSE) on the test data.

Task 9 - 3 Marks

Create scatter plots comparing the predicted values to the ground truth for the two output variables, y_1 and y_2 .

Submission Guidelines

- The content that you submit must be your individual work.
- Submit both .py and .ipynb file formats. Both submissions are mandatory for credit.
- Name the files as "<roll_number>_assignment_1.py" and "<roll_number>_assignment_1.ipynb". Submit them in a single zip file named "<roll_number>_assignment_1.zip". For example, if your roll number is 23AI91R01, the submission file should be named 23AI91R01_assignment_1.zip (Note: do not submit .rar files).

- Ensure all code is well-commented and explanations are included using text cells in the Jupyter Notebook.
- Submit your assignment through Moodle before the deadline. Late submissions or email submissions will not be accepted.

TA for this assignment: Raviteja

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